

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Areas Related to Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sector and segment of a circle

1. A sector of a circle of radius 21 cm has a central angle of 60° . Find the length of its arc. (a) 22 cm (b) 44 cm (c) 88 cm (d) 11 cm **CBSE 2026, 1 mark**

SOLUTION

$$\text{Arc length} = \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$

Answer: (a)

2. The hour hand of a clock is 7 cm long. Through what angle does it turn between 7:00 a.m. and 8:10 a.m.? (a) $\left(\frac{35}{4}\right)^\circ$ (b) $\left(\frac{35}{2}\right)^\circ$ (c) 35° (d) 70° **CBSE 2026, 1 mark**

SOLUTION

The hour hand turns 360° in 12 hours, which is 30° in an hour or $\frac{1}{2}^\circ$ in a minute.

From 7:00 to 8:10 is 70 minutes: $70 \times \frac{1}{2} = 35^\circ$. (The length of the hand is not needed.)

Answer: (c)

3. A circular ornament of diameter 35 cm is made from silver wire. Wire is also used for 5 diameters, which cut the circle into 10 equal sectors. (i) Find the radius. (ii) Find the circumference. (iii) Find the total length of wire needed, OR the area of each sector. **CBSE 2026, 4 marks**

SOLUTION

(i) Radius = $\frac{35}{2}$. **Radius = 17.5 cm.**

(ii) Circumference = $2 \times \frac{22}{7} \times \frac{35}{2}$. **Circumference = 110 cm.**

(iii) Wire = circumference + 5 diameters = $110 + 5 \times 35$. **Total wire = 285 cm.**

OR: area of the circle = $\frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = 962.5 \text{ cm}^2$, and each sector is one-tenth of it. **Area of each sector = 96.25 cm².**

4. A sector with a central angle of 72° has an area of 40π square units. Find the radius of the circle.

CBSE 2025, 1 mark

SOLUTION

$$\frac{72}{360} \times \pi r^2 = 40\pi, \text{ so } \frac{r^2}{5} = 40$$

$$r^2 = 200$$

Answer: $r = 10\sqrt{2}$ units

5. A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? CBSE 2025, 1 mark

SOLUTION

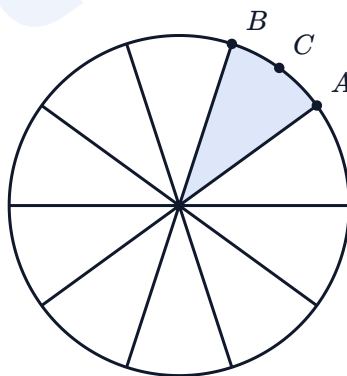
$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r, \text{ so } 20 = \frac{\theta}{360} \times 2\pi \times \frac{60}{\pi} = \frac{\theta}{360} \times 120$$

$$\theta = \frac{20 \times 360}{120}$$

Answer: 60°

6. A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. A and B are the ends of one sector on the circle and C lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc ACB . (iii) Find the area of each sector, OR the total length of wire used.

CBSE 2025, 4 marks



SOLUTION

(i) Ten equal sectors share 360° . **Each central angle is 36° .**

(ii) Radius $r = \frac{35}{2}$ mm, so the circumference is $2 \times \frac{22}{7} \times \frac{35}{2} = 110$ mm. Arc $ACB = \frac{36}{360} \times 110$.
Arc $ACB = 11$ mm.

(iii) Area of the circle = $\frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = 962.5$ mm². Each sector is one-tenth of it. **Area of each sector = 96.25 mm².**

OR: wire used = circumference + 5 diameters = $110 + 5 \times 35$. **Total wire = 285 mm.**

7. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. **CBSE 2024, 5 marks**

SOLUTION

$$(i) \text{ Length of arc} = \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22 \text{ cm}$$

$$(ii) \text{ Area of sector} = \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231 \text{ cm}^2$$

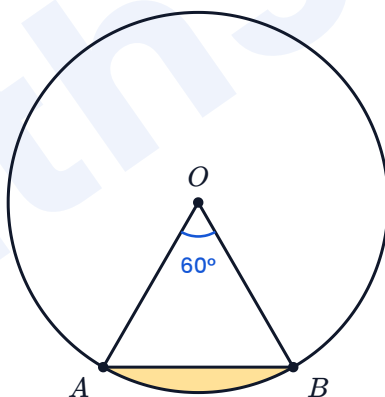
The triangle formed by the two radii and the chord has an angle of 60° between equal sides, so it is equilateral with side 21 cm. Its area = $\frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4} \text{ cm}^2$

$$\text{Area of minor segment} = \left(231 - \frac{441\sqrt{3}}{4} \right) \text{ cm}^2 \approx 231 - 190.95 = 40.05 \text{ cm}^2 \text{ (taking } \sqrt{3} = 1.732 \text{)}$$

$$\text{Answer: (i) 22 cm (ii) } \left(231 - \frac{441\sqrt{3}}{4} \right) \text{ cm}^2 \approx 40.05 \text{ cm}^2$$

8. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the minor segment and the area of the major segment. Take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$.

CBSE 2023, 5 marks



SOLUTION

$$\text{Area of the sector} = \frac{60}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{616}{6} \approx 102.67 \text{ cm}^2$$

The triangle formed by the two radii and the chord has a 60° angle between equal sides, so it is equilateral with side 14 cm. Its area is $\frac{\sqrt{3}}{4} \times 14^2 = 49\sqrt{3} \approx 84.87 \text{ cm}^2$.

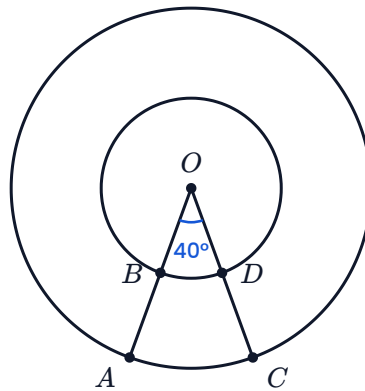
$$\text{Minor segment} = 102.67 - 84.87 = 17.80 \text{ cm}^2$$

$$\text{Area of the circle} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2, \text{ so the major segment} = 616 - 17.80 = 598.20 \text{ cm}^2.$$

$$\text{Answer: minor segment} \approx 17.80 \text{ cm}^2; \text{ major segment} \approx 598.20 \text{ cm}^2$$

9. Two concentric circles with centre O have radii 7 cm and 14 cm. A sector of angle 40° is cut away from the ring between them. Find the area of the part of the ring that remains. Take $\pi = \frac{22}{7}$.

CBSE 2016, 3 marks



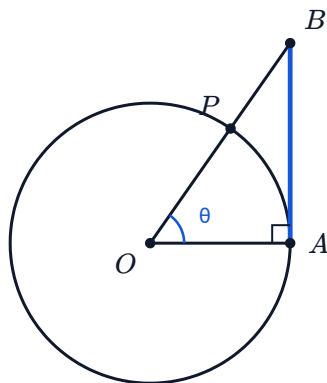
SOLUTION

The remaining part of the ring corresponds to an angle of $360^\circ - 40^\circ = 320^\circ$.

$$\begin{aligned} \text{Area} &= \frac{320}{360} \times \pi(R^2 - r^2) = \frac{8}{9} \times \frac{22}{7} \times (14^2 - 7^2) = \frac{8}{9} \times \frac{22}{7} \times 147 \\ &= \frac{8}{9} \times 462 = \frac{3696}{9} \end{aligned}$$

Answer: about 410.67 cm²

- 10.** OAP is a sector of a circle with centre O , radius r and angle θ (in degrees). AB is perpendicular to the radius OA and meets OP produced at B . Prove that the perimeter of the region bounded by AB , BP and the arc PA is $r \left[\tan \theta + \sec \theta + \frac{\pi \theta}{180} - 1 \right]$. **CBSE 2016, 4 marks**



SOLUTION

In right triangle OAB : $\tan \theta = \frac{AB}{r}$, so $AB = r \tan \theta$, and $\sec \theta = \frac{OB}{r}$, so $OB = r \sec \theta$.

$$BP = OB - OP = r \sec \theta - r$$

$$\text{Arc } PA = \frac{\theta}{360} \times 2\pi r = \frac{\pi r \theta}{180}$$

$$\text{Perimeter} = AB + BP + \text{arc } PA = r \tan \theta + r \sec \theta - r + \frac{\pi r \theta}{180}$$

$$= r \left[\tan \theta + \sec \theta + \frac{\pi \theta}{180} - 1 \right]. \text{ Proved.}$$

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